

Unit 3, Lesson 1: Powers of 10

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- ☐ I can use the Laws of Exponents to evaluate numerical expressions with a base of 10 and exponents greater than or equal to 0.

3.1.1 Warm-Up: Grains of Sand

The beaches on Siesta Key are 99% quartz, meaning that the beaches are made of some of the finest grains of sand.



1. Suppose you have a teaspoon of the sand from Siesta Key. How could the teaspoon of sand be used to estimate the number of grains of sand in a bucket of sand?

A group of researchers at the University of Hawaii used a teaspoon of sand to estimate the number of grains of sand on all of the beaches on the Earth. Their estimate was that there are roughly seven quintillion, five hundred quadrillion grains of sand. A quintillion is 1,000,000,000,000,000,000 and a quadrillion is 1,000,000,000,000,000.

2. Write seven quintillion, five hundred quadrillion.

3.1.2 Exploration: Powers of Ten

1. Use a scientific calculator to find the value of each power of 10.

Power of 10	Value
10^1	
10^2	
10^3	
10^4	
10^5	
10^6	
10^7	
10^8	
10^9	
10^{10}	

2. What do you notice about the relationship between the power of 10 and its value?

3. Work with your partner to complete the table.

Power of 10	Value	Place Value of 1
10^1	10	Tens
10^2		
10^3		
10^4		
10^5		
10^6		

4. Work with your partner to describe how the exponent of the power of ten relates to place value.
5. Based on the pattern in the table, what is the value of 10^0 ?

3.1.3 Try It: Make a Guess

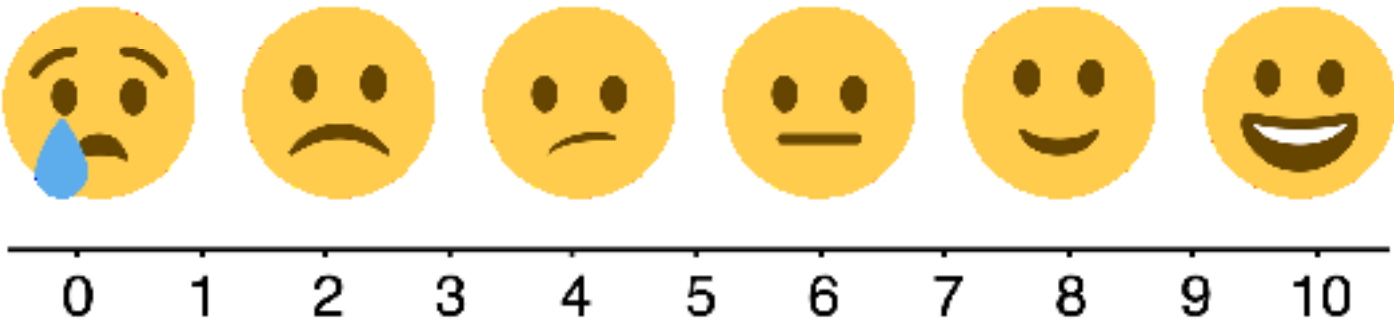
1. Without your partner or a calculator, make a guess on the value of each mathematical statement. Do not worry about being wrong. Remember, mistakes are a part of learning.

Whenever possible, write your guess using a power of 10. Show your work or explain your thinking.

Mathematical Statement	My Guess	My Work or Explanation
$10^6 + 10^7$		
$10^2 \times 10^6$		
$10^5 - 10^3$		
$\frac{10^6}{10^4}$		
$(10^4)^2$		
$\frac{1}{10^3}$		

2. How confident are you in your guesses?

Use the emoji scale to rate your confidence in your guesses.



3.1.4 Exploration: Using a Calculator to Check

1. Use a calculator to complete the following.
- a. Find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10.

Mathematical Statement	Value Found Using the Calculator	Value Written as a Power of 10
$10^6 + 10^7$		
$10^2 \times 10^6$		
$10^5 - 10^3$		
$\frac{10^6}{10^3}$		
$(10^4)^2$		
$\frac{1}{10^2}$		
- b. Compare your guesses from the Try It to the values you found for each expression using a calculator.

c. Discuss with your partner which mathematical statements could be rewritten using a power of 10 and which could not be. Summarize your findings.

3.1.5 Refresher: Laws of Exponents

The Laws of Exponents may help explain why some of the mathematical statements could be rewritten using a power of 10.



Product of Powers Exponent Law

When multiplying two powers that have the same nonzero base, keep the base and add the exponents.

1. Which is an example of the product of powers exponent law?

- ☐ $(a^m)^n = a^{mn}$
- ☐ $(a^m)^n = a^{m+n}$
- ☐ $a^m \cdot a^n = a^{mn}$
- ☐ $a^m \cdot a^n = a^{m+n}$



Power of a Power Exponent Law

When raising a power to a power, keep the base and multiply the exponents.

2. Which is an example of the power of a power exponent law?

- ☐ $(a^m)^n = a^{mn}$
- ☐ $(a^m)^n = a^{m+n}$
- ☐ $a^m \cdot a^n = a^{mn}$
- ☐ $a^m \cdot a^n = a^{m+n}$



Quotient of Powers Exponent Law

When dividing two powers that have the same nonzero base, keep the base and subtract the exponents.

3. Which is an example of the quotient of powers exponent law?

☐ $\frac{a^m}{a^n} = a^{m-n}$

☐ $\frac{a^m}{a^n} = (a - a)^{m-n}$

☐ $\frac{a^m}{a^n} = 1^{m-n}$



Identity Exponent Law

When raising a base that is a nonzero real number to the power of 1, the base is a factor one time, thus, has the same value as the base.

4. Write an example of the identity exponent law.



Zero Exponent Law

When a nonzero real number is raised to the zero power, a^0 , its value equals 1.

5. Write an example of the zero exponent law.

3.1.6 Guided Practice: Aligning Powers of 10 Statements with the Laws of Exponents

1. Determine which of the mathematical statements align with a law of exponents. Write the law in the table. If no law applies, then write "none."

Mathematical Statement	Law of Exponents
$10^6 + 10^7$	
$10^2 \times 10^5$	
$10^5 - 10^3$	
$\frac{10^6}{10^4}$	
$(10^4)^2$	
$\frac{1}{10^2}$	

2. Numbers like a quintillion, 1,000,000,000,000,000,000, and a quadrillion, 1,000,000,000,000,000, are often used by scientists. When using such large numbers, scientists have developed a shortened method of writing these numbers called scientific notation. Scientific notation uses the powers of 10 to represent these numbers.

a. Write 1,000,000,000,000,000,000 using a power of 10.

b. Write 1,000,000,000,000,000 using a power of 10.

3.1.7 Wrap-Up: Reflection

1. Choose the question and sign that best describes your understanding of today's lesson.

Are you confused?	Are you in the school zone?	Are you ready for the highway?
		

2. Briefly explain why you feel this way.

